

Protein

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Introduction to Proteins

Proteins are fundamental components essential to all organisms. They are biological macromolecules composed of one or more linear chains of amino acids linked in a specific sequence. Proteins govern the complex functions and structure of cells, tissues, and organs, each performing a distinct role.

Functions and Diversity

Proteins exhibit enormous structural complexity and functional sophistication. Major categories include:

- **Enzymes:** Catalyze biochemical reactions, forming metabolism.
- **Structural Proteins:** Provide mechanical support, e.g., collagen, keratin.
- **Transport Proteins:** Carry molecules or ions, e.g., hemoglobin.
- **Motor Proteins:** Generate movement, e.g., myosin, kinesin.
- **Signal/Receptor Proteins:** Hormones or receptors for communication.
- **Antibodies:** Protect against foreign molecules.
- **Gene Regulatory Proteins:** Switch genes on/off.

Synthesis and Abundance

Protein synthesis follows the central dogma: DNA $\rightarrow$ RNA $\rightarrow$ Protein. Proteins account for 15% of a cell's dry mass, ranging from 30 to over 10,000 amino acids, typically 50–2000. Characteristic size is 3–10 nm.

Proteins in the Body

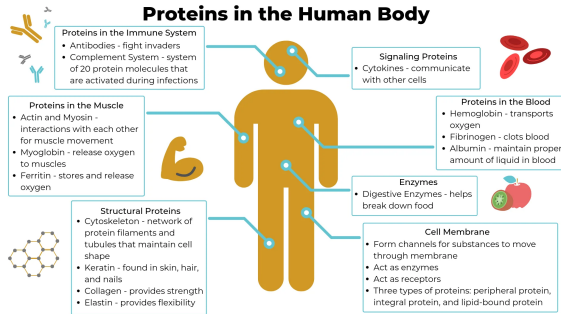


Figure: Proteins as essential components of the human body, highlighting their abundance and roles.

Source: nusep.eu

Types of Proteins by Function

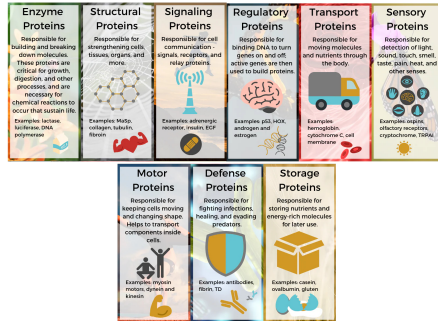


Figure: Illustration of major protein categories and their functions.

Source: nusep.eu

Amino Acids (The Monomers)

Proteins are polymers built from amino acids. All amino acids share:

- -carbon linked to an amino group and a carboxyl group.
- Unique side chain (R group) determining chemical identity.

There are 20 common amino acids across organisms.

Amino Acid Structure

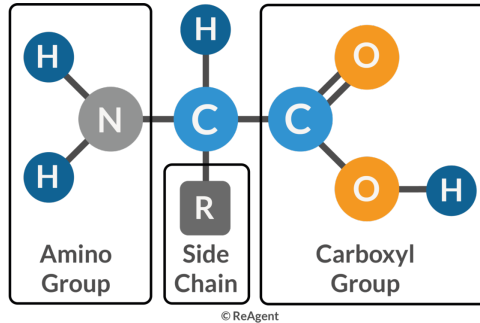


Figure: Amino acids consist of a central carbon, amino group, carboxyl group, and side chain.

Source: reagent.co.uk

Peptide Bonds

Amino acids link via peptide bonds:

- Covalent bond connects carbonyl of one amino acid to amino group of next.
- Formed via condensation reaction (water released).
- Polypeptide chain has polarity: N-terminus and C-terminus.

Peptide Bond Formation

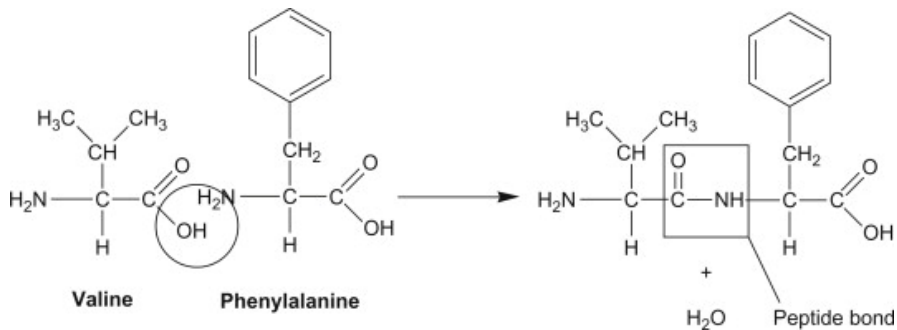


Figure: Condensation reaction forming peptide bonds between amino acids.

Source: sciencedirect.com

Types of Amino Acids

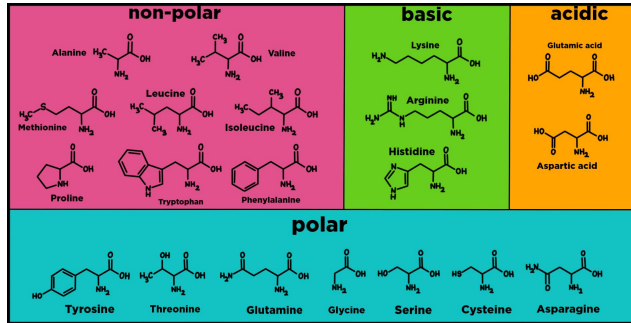


Figure: Twenty amino acids categorized by side chain properties.

Source: expii.com

Folding Principle and Stabilizing Forces

Protein folding is guided by amino acid sequence:

- Achieves 3D conformation of minimal free energy.
- Stabilized by hydrogen bonds, hydrophobic interactions, ionic attractions, van der Waals forces.
- Disulfide bonds provide additional stability in extracellular proteins.

Orders of Protein Structure

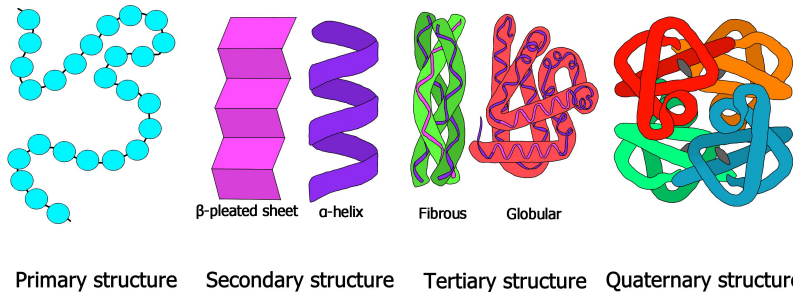


Figure: Secondary (alpha helices and beta sheets), tertiary, and quaternary structures.

Source: chembam.com

Protein Domains

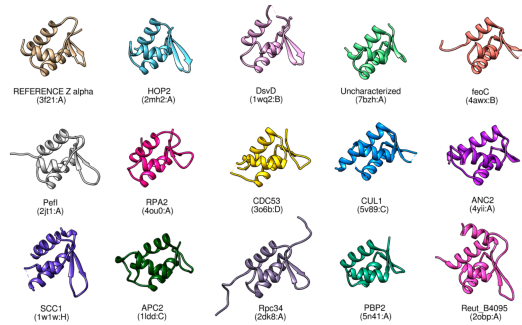


Figure: Domains: independently folding units in larger proteins.

Source: M. Bartas, K. Slychko, V. Brázda, J. Červeň, C. A. Beaudoin, T. L. Blundell, P. Pečinka, IJMS 2022, 23, 768.

Bonds Stabilizing Protein Structures

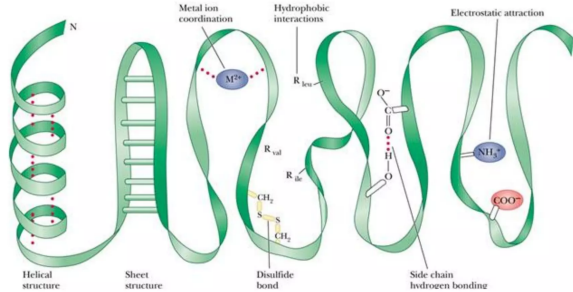


Figure: Stabilizing forces.

Source: slideshare.net

What is Protein Folding?

Protein folding is the physical process by which a linear polypeptide chain acquires its specific three-dimensional structure.

- Driven by the amino acid sequence (*Anfinsen's dogma*).
- Occurs co-translationally (during synthesis) or post-translationally.
- Proper folding is essential for correct biological function.

Forces Driving Folding

Folding is a thermodynamically favorable process leading to a low-energy native state, stabilized by:

- **Hydrophobic interactions:** Nonpolar residues buried inside.
- **Hydrogen bonds:** Stabilize secondary structures.
- **Ionic interactions:** Between charged side chains.
- **van der Waals forces:** Compact packing of atoms.
- **Disulfide bonds:** Covalent bridges adding stability.

Chaperones and Folding Assistance

Molecular chaperones assist protein folding by preventing aggregation or misfolding:

- **Hsp70:** Binds nascent chains, preventing premature folding.
- **Chaperonins (GroEL/GroES):** Provide isolated folding chambers.
- **Hsp90:** Assists in late folding and signal transduction proteins.

Note: Chaperones do not determine the final structure—they help reach it efficiently.

Protein Misfolding and Diseases

Improperly folded proteins can aggregate and cause diseases:

- **Alzheimer's disease:** Amyloid- peptide aggregation.
- **Parkinson's disease:** Misfolded -synuclein.
- **Prion diseases:** Infectious misfolded prion proteins.

Protein homeostasis (“proteostasis”) networks monitor folding, refolding, and degradation.

Stability and Denaturation

Protein stability depends on temperature, pH, and solvent conditions.

- **Denaturation:** Loss of native conformation, usually reversible only under mild conditions.
- **Agents:** Heat, urea, guanidinium chloride, detergents.
- **Thermostable proteins:** Found in extremophiles; stabilized by salt bridges and compact cores.

Protein Function Overview

Proteins perform a vast range of biological activities that are essential to life. Their specific function arises from their structure and binding interactions.

- Each protein has a unique 3D shape determining its activity.
- Even minor mutations may alter or abolish function.
- Functions can be inferred from conserved motifs or structures.

Major Categories of Protein Functions

- **Catalysis:** Enzymes accelerate biochemical reactions.
- **Structural Support:** Collagen, keratin provide rigidity.
- **Transport and Storage:** Hemoglobin, ferritin, albumin.
- **Signaling:** Hormones and receptors transmit information.
- **Defense:** Antibodies and complement proteins protect cells.
- **Regulation:** Transcription factors and kinases control activity.
- **Movement:** Myosin and actin enable cellular motion.

Examples of Key Functional Proteins

Protein	Function	Example Organism
Hemoglobin	Oxygen transport	Humans
Myosin	Muscle contraction	Animals
DNA Polymerase	DNA synthesis	All cells
Antibody (IgG)	Immune defense	Vertebrates
Insulin	Glucose regulation	Mammals
GFP	Fluorescent marker	<i>Aequorea victoria</i>

Table: Representative proteins and their biological roles.

Bioinformatics Approaches to Protein Function

Functional prediction is a major goal in bioinformatics:

- **Sequence-based:** Motif and domain search (Pfam, PROSITE).
- **Structure-based:** 3D modeling and docking (PDB, AlphaFold).
- **Evolutionary:** Homology and phylogenetic profiling.
- **Machine learning:** Predict function from embeddings or structure graphs.

Databases like UniProt and InterPro provide curated annotations and ontology terms.

What Are Protein Databases?

Protein databases store and organize experimental and computational data about proteins. They are indispensable for:

- Sequence retrieval and comparison.
- Functional annotation.
- Structural modeling.
- Drug discovery and target identification.

Primary vs Secondary Databases

- **Primary Databases:** Contain experimentally determined data.
 - *UniProtKB, PDB, GenPept.*
- **Secondary Databases:** Derive information from primary data (annotations, motifs, families).
 - *Pfam, InterPro, PROSITE, SMART.*

Major Protein Databases

- **UniProtKB:** Comprehensive protein sequence and functional information.
- **PDB (Protein Data Bank):** 3D structures from X-ray, NMR, and cryo-EM.
- **InterPro:** Integrates multiple signature databases for domain analysis.
- **Pfam:** Protein families and hidden Markov models (HMMs).
- **PROSITE:** Database of motifs and patterns.
- **AlphaFold DB:** AI-predicted structures of millions of proteins.

Example: UniProt Entry

The screenshot displays the UniProtKB entry for P05067 (A4_HUMAN). The top navigation bar includes links for BLAST, Align, Peptide search, ID mapping, SPARQL, and UniProtKB. The main header shows the protein name and accession number. Below this, a table provides key information:

Category	Value
Protein ⁱ	Amyloid-beta precursor protein
Gene ⁱ	APP
Status ⁱ	UniProtKB reviewed (Swiss-Prot)
Organism ⁱ	Homo sapiens (Human)
Amino acids	770 (go to sequence)
Protein existence ⁱ	Evidence at protein level
Annotation score ⁱ	5/5

Below the table, there are tabs for Entry, Variant viewer (880), Feature viewer, Genomic coordinates, Publications, External links, and History. A 'Tools' section includes links for Download, Add, Add a publication, and Entry feedback.

The 'Function' section is expanded, showing the following text:

Functionⁱ

Functions as a cell surface receptor and performs physiological functions on the surface of neurons relevant to neurite growth, neuronal adhesion and axonogenesis. Interaction between APP molecules on neighboring cells promotes synaptogenesis (PubMed:25122912). Involved in cell mobility and transcription regulation through protein-protein interactions. Can promote transcription activation through binding to APBB1-KAT5 and inhibits Notch signaling through interaction with Numb. Couples to apoptosis-inducing pathways such as those mediated by G(o) and JIP. Inhibits G(o) alpha ATPase activity (By similarity). Acts as a kinesin I membrane receptor, mediating the axonal transport of beta-secretase and presenilin 1 (By similarity). By acting as a kinesin I membrane receptor, plays a role in axonal anterograde transport of cargo towards synapses in axons (PubMed:17062754, PubMed:23011729). Involved in copper homeostasis/oxidative stress through copper ion reduction. In vitro, copper-metallated APP induces neuronal death directly or is potentiated by G234.

Figure: Example of a UniProtKB protein entry showing sequence, function, and domain information.

Integrating Data from Multiple Sources

Protein databases are often cross-linked:

- UniProt $\leftrightarrow$ PDB (structure links).
- UniProt $\leftrightarrow$ Pfam/InterPro (domains).
- UniProt $\leftrightarrow$ GO (Gene Ontology annotations).

Goal: Enable unified access to multi-dimensional data — sequence, structure, and function.

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The End

Questions? Comments?